

RIFAT ERDEM SAHIN

AI Engineering Lead | LLM Solutions Architect

** Location:** London, United Kingdom ** Citizenship:** British ** Email:**
contact@rifaterdemsahin.com ** Phone:** +44 7848 024173 ** LinkedIn:**
linkedin.com/in/rifaterdemsahin ** GitHub:** github.com/rifaterdemsahin **
Portfolio:** <https://rifaterdemsahin.com> ** Schedule a Call:** <https://calendly.com/rifaterdem/schedule>

PROFESSIONAL SUMMARY

AI Engineering Lead specializing in **Generative AI, Large Language Models, and production-grade AI systems**. Deep hands-on experience architecting and deploying enterprise LLM applications with focus on **RAG architectures, AI security, and scalable deployment**. Proven track record building AI-powered automation systems that delivered 300% productivity improvements and 30% cost reductions. Expert in bridging the gap between cutting-edge AI research and production-ready, secure, enterprise solutions.

CORE AI COMPETENCIES

Generative AI & LLM Engineering - Deep expertise building and deploying production Generative AI solutions - Extensive hands-on experience developing Large Language Model applications - Advanced prompt engineering, fine-tuning, and model optimization - Expert in LangChain, LlamaIndex, and modern AI orchestration frameworks

RAG Architecture & Vector Systems - Advanced implementation of Retrieval Augmented Generation (RAG) architectures - Expert in vector databases (Pinecone, Weaviate, ChromaDB, FAISS) - Document processing pipelines and semantic search optimization - Hybrid search strategies combining dense and sparse retrieval

AI Security & Governance - Expert-level knowledge of AI security practices and threat models - Prompt injection prevention and adversarial attack mitigation - Data privacy controls, PII detection, and secure model deployment - Compliance frameworks for regulated industries (finance, healthcare)

AI System Architecture - Track record shaping enterprise AI architecture decisions - Scalable, maintainable AI solutions with proper observability - MLOps pipelines for continuous model deployment and monitoring - Cost optimization strategies for LLM inference at scale

AI Testing & Quality Assurance - Comprehensive testing approaches: unit, integration, end-to-end - LLM evaluation frameworks and metrics (RAGAS, LangSmith) - A/B testing strategies for prompt and model optimization - Automated quality gates for AI system releases

KEY AI ACCOMPLISHMENTS

2024 | Goldman Sachs | Muscat, Oman

AI-Driven CI/CD Automation Framework - **Challenge:** Manual deployment processes causing bottlenecks and errors in enterprise fintech environment - **Solution:** Architected and deployed AI-powered CI/CD framework using LLMs for intelligent automation - **Impact:** - 300% increase in deployment frequency - 30% reduction in operational costs - 85% reduction in deployment-related incidents - **Technologies:** Python, LangChain, OpenAI GPT-4, Azure OpenAI, Kubernetes, GitLab CI/CD - **AI Components:** Custom RAG system for documentation, automated code review agents, intelligent error detection

2023 | Ypsomed | Switzerland

IoT Data Intelligence & Predictive Systems - **Challenge:** Complex IoT device data requiring intelligent analysis and predictive maintenance - **Solution:** Implemented ML-powered analytics pipeline with automated anomaly detection - **Impact:** - 40% reduction in versioning conflicts through intelligent merge strategies - Predictive maintenance reducing device failures by 35% - **Technologies:** Python, TensorFlow, Azure ML, IoT Hub, Time-series analysis - **AI Components:** Anomaly detection models, predictive maintenance algorithms, automated data quality checks

2022 | Cushman & Wakefield | London, UK

Intelligent ETL & Data Pipeline Automation - **Challenge:** Manual data processing creating bottlenecks in real estate analytics - **Solution:** Built AI-enhanced ETL pipelines with intelligent data validation and transformation - **Impact:** - 50% increase in data processing speed - 90% reduction in data quality issues - Real-time analytics enabling faster business decisions - **Technologies:** Python, Apache Airflow, Azure Data Factory, ML-based data validation - **AI Components:** Automated data quality scoring, intelligent schema mapping, anomaly detection in data flows

TECHNICAL EXPERTISE

AI/ML Frameworks & Tools

LLM Platforms: OpenAI API (GPT-4, GPT-3.5), Anthropic Claude, Azure OpenAI Service, AWS Bedrock **Orchestration:** LangChain, LlamaIndex, Semantic Kernel, Haystack **Vector Databases:** Pinecone, Weaviate, ChromaDB, FAISS, Qdrant, Milvus **ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers **Fine-tuning:** LoRA, QLoRA, PEFT, Supervised Fine-Tuning (SFT) **Evaluation:** RAGAS, LangSmith, Weights & Biases, MLflow

AI Development & MLOps

Languages: Python (Advanced), JavaScript/TypeScript, Bash, SQL **AI Libraries:** OpenAI Python SDK, LangChain, pandas, numpy, sentence-transformers **MLOps:** MLflow, Kubeflow, Azure ML, AWS SageMaker, Model versioning **Experiment Tracking:** Weights & Biases, MLflow, TensorBoard **Data Processing:** Apache Spark, Pandas, Dask, Apache Airflow

Cloud & Infrastructure

Cloud Platforms: Azure (Azure OpenAI, ML, Functions), AWS (Bedrock, Lambda, SageMaker), GCP **Containers:** Docker, Kubernetes, Helm (for ML workloads) **Infrastructure as Code:** Terraform, Ansible **CI/CD:** Jenkins, GitLab CI, GitHub Actions, Azure DevOps **Monitoring:** Prometheus, Grafana, ELK Stack, Application Insights

AI Security & Compliance

Security Clearances: UK SC (valid until 2028), NATO (valid until 2029) **AI Security:** Prompt injection prevention, adversarial robustness, model security **Data Privacy:** PII detection/redaction, data anonymization, GDPR compliance **Governance:** Model governance frameworks, AI ethics, bias detection

NOTABLE AI PROJECTS & IMPLEMENTATIONS

Production RAG Systems

- Built enterprise-grade RAG systems processing 10M+ documents
- Implemented hybrid search combining semantic and keyword search
- Optimized retrieval pipelines reducing latency by 60%
- Custom chunking strategies improving retrieval accuracy by 40%

AI Agent Development

- Designed multi-agent systems for complex workflow automation
- Implemented ReAct and function-calling patterns for tool use

- Built autonomous agents with memory and planning capabilities
- Created agent orchestration frameworks for parallel execution

LLM Fine-tuning & Optimization

- Fine-tuned domain-specific models using LoRA/QLoRA
- Optimized prompts achieving 35% cost reduction while improving accuracy
- Implemented caching strategies reducing API costs by 50%
- Model quantization for edge deployment scenarios

AI Security Implementations

- Built prompt injection detection systems with 95% accuracy
- Implemented PII detection/redaction pipelines for sensitive data
- Created guardrails preventing model misuse and hallucinations
- Developed secure model serving infrastructure with encryption at rest/transit

PROFESSIONAL EXPERIENCE HIGHLIGHTS

Senior AI Engineer / DevOps Lead | 2020 - Present *Goldman Sachs, Ypsomed, Cushman & Wakefield*

- Led AI transformation initiatives across finance, healthcare, and real estate sectors
- Architected production LLM systems serving millions of requests monthly
- Implemented MLOps pipelines reducing model deployment time from weeks to hours
- Mentored engineering teams on AI best practices and secure deployment patterns
- Established AI governance frameworks ensuring compliance and ethical AI use

Enterprise Transformation Architect | 2016 - 2020 *Microsoft, Emerson, Various Fortune 500*

- Led digital transformation initiatives incorporating AI/ML capabilities
 - Built ML-powered automation systems increasing operational efficiency by 50%
 - Designed cloud-native architectures supporting ML workloads at scale
 - Evangelized AI adoption through workshops, training, and proof-of-concepts
-

EDUCATION

Bachelor of Science Southern New Hampshire University, USA | 2013

CERTIFICATIONS & CONTINUOUS LEARNING

Microsoft Certified Architect in Cloud Solutions (70-532) Azure AI Engineer Associate AWS Machine Learning Specialty Certified Kubernetes Administrator (CKA)

Continuous Learning: - Active contributor to AI/ML open-source projects - Regular participant in AI research paper discussions - Following latest developments in LLM research (GPT, Claude, Llama, Mistral) - Experimenting with cutting-edge AI techniques and frameworks

SPEAKING & THOUGHT LEADERSHIP

- Technical blog on AI engineering best practices at rifaterdemsahin.com
 - Internal tech talks on RAG architectures and LLM security
 - Mentorship in AI/ML communities and engineering teams
 - Code examples and tutorials shared via GitHub
-

SECURITY CLEARANCES

UK SC (Security Check) - Valid until 2028 **NATO Clearance** - Valid until 2029 **Background Checks:** Watchdog (2024), Sterling (2019)

AVAILABILITY & CONTACT

Immediate Availability for contract or permanent AI engineering roles

Schedule a Discussion: <https://calendly.com/rifaterdem/schedule> **Email:** contact@rifaterdemsahin.com **Phone:** +44 7848 024173

SUPPORTING DOCUMENTS

Download Full CV (PDF): https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv_summary_2025_may.pdf

Download Full CV (Word): https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv_summary_2025_may.docx

Technical Portfolio & Presentations: <https://rifaterdemsahin.com/wp-content/uploads/2025/02/rifaterdemsahinprofilepresentation.v2025.2.pdf>

References and detailed project portfolios available upon request